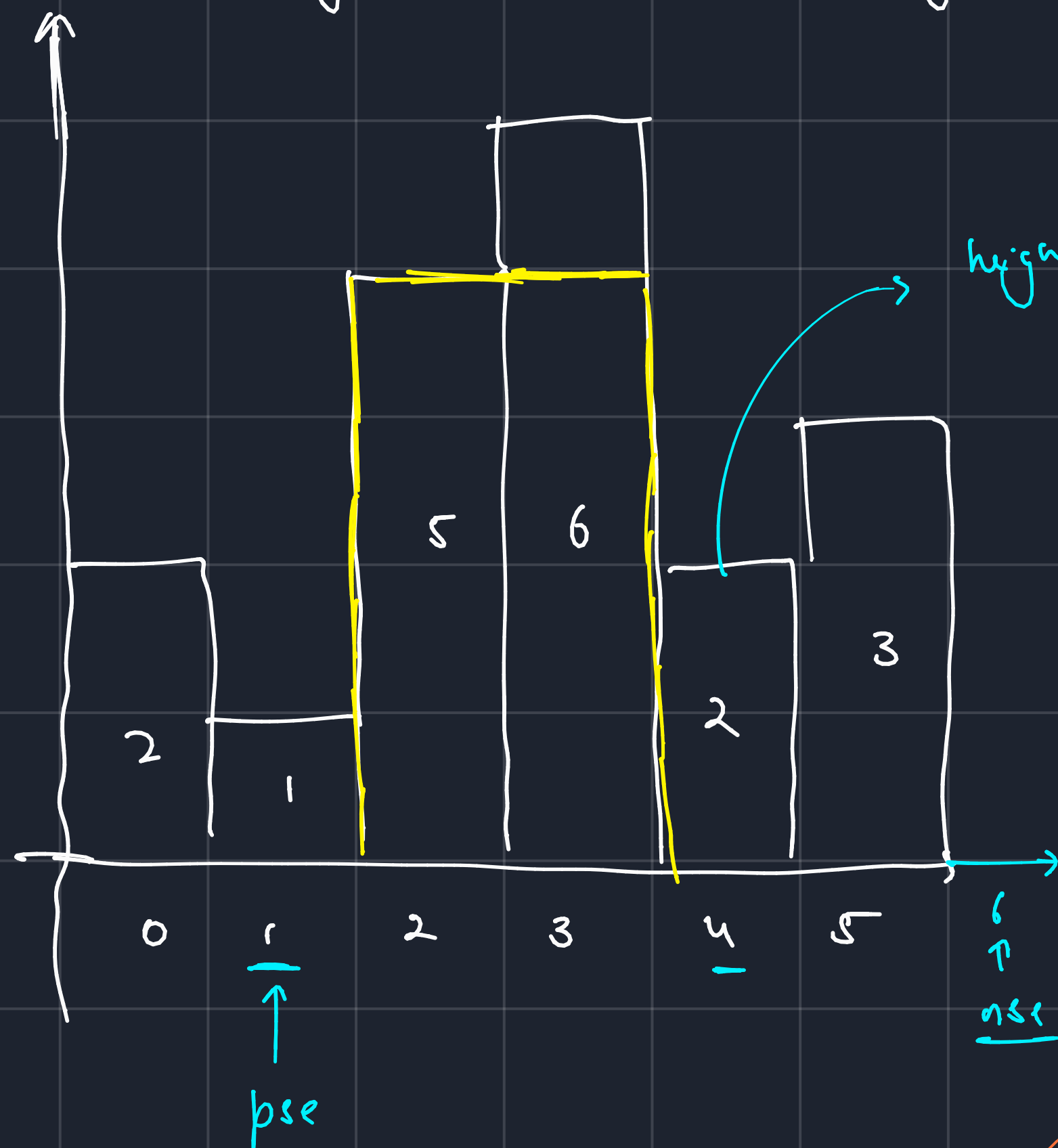


↳ largest rectangle in histogram.



↳ This is the brute force

↳ pse, nse

↳ Time complexity $\rightarrow T.C \rightarrow 2 \times O(2N)$

↳ $O(4N)$ $\nearrow$ maxi loop

total $\rightarrow O(5N)$ ($1N$)

↳ stack

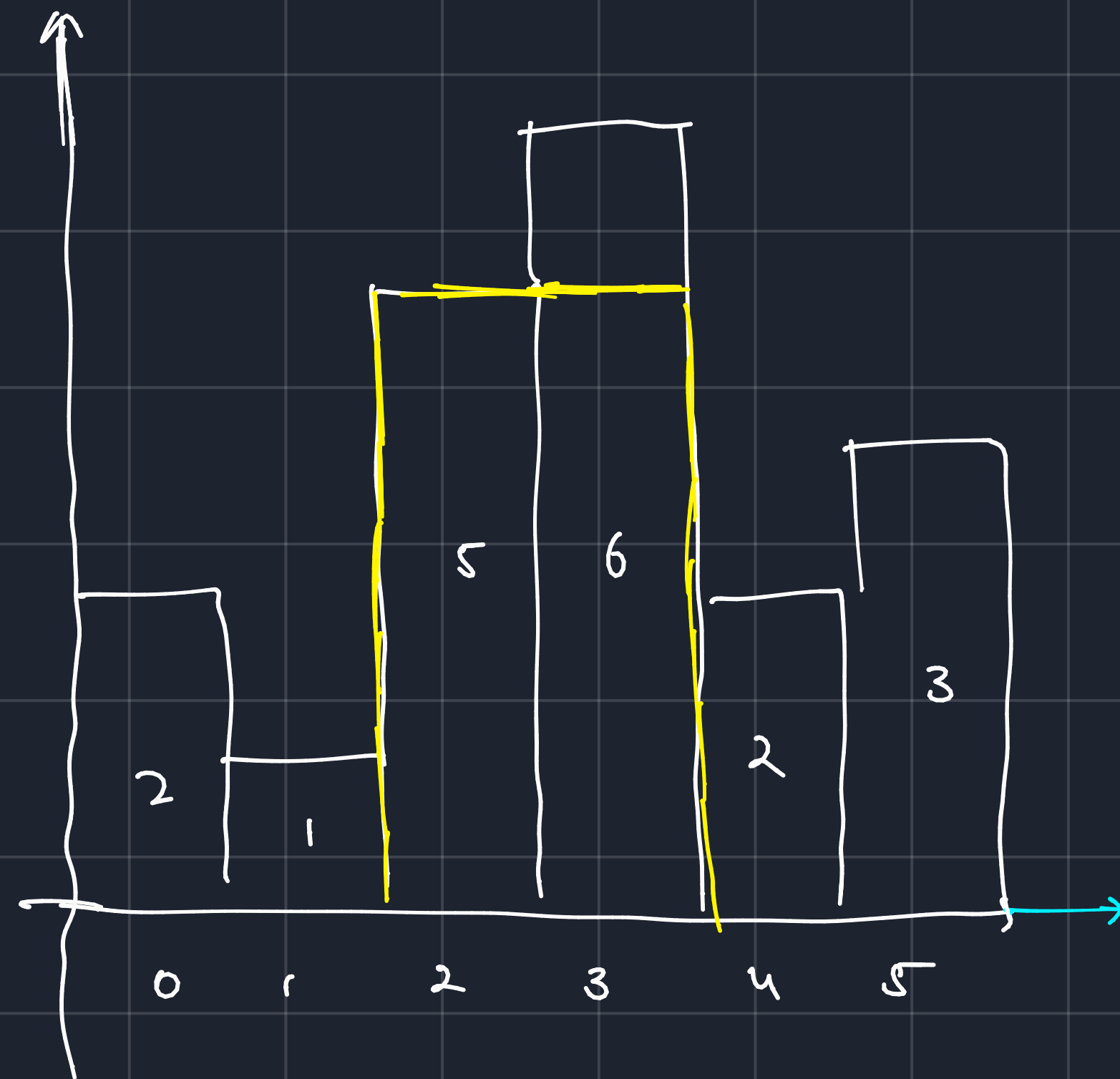
size of
the
array

↳ if there is no nse $\rightarrow$ assign n
↳ if there is no pse assign $= 1$.

↳ $arr[i] \times (nse - pse - 1)$

↳ area of a rectangle

↳ single pass



↳ Brute

$$\text{arr}[i] \times (\text{n} - \text{pre} - 1)$$

we will have
to traverse
the array once

we need to compute this
on the fly.

↳ we can compute pre on the fly as we
are traversing in the direction.

↳ but what about n-se- → we are yet to see
the next elements

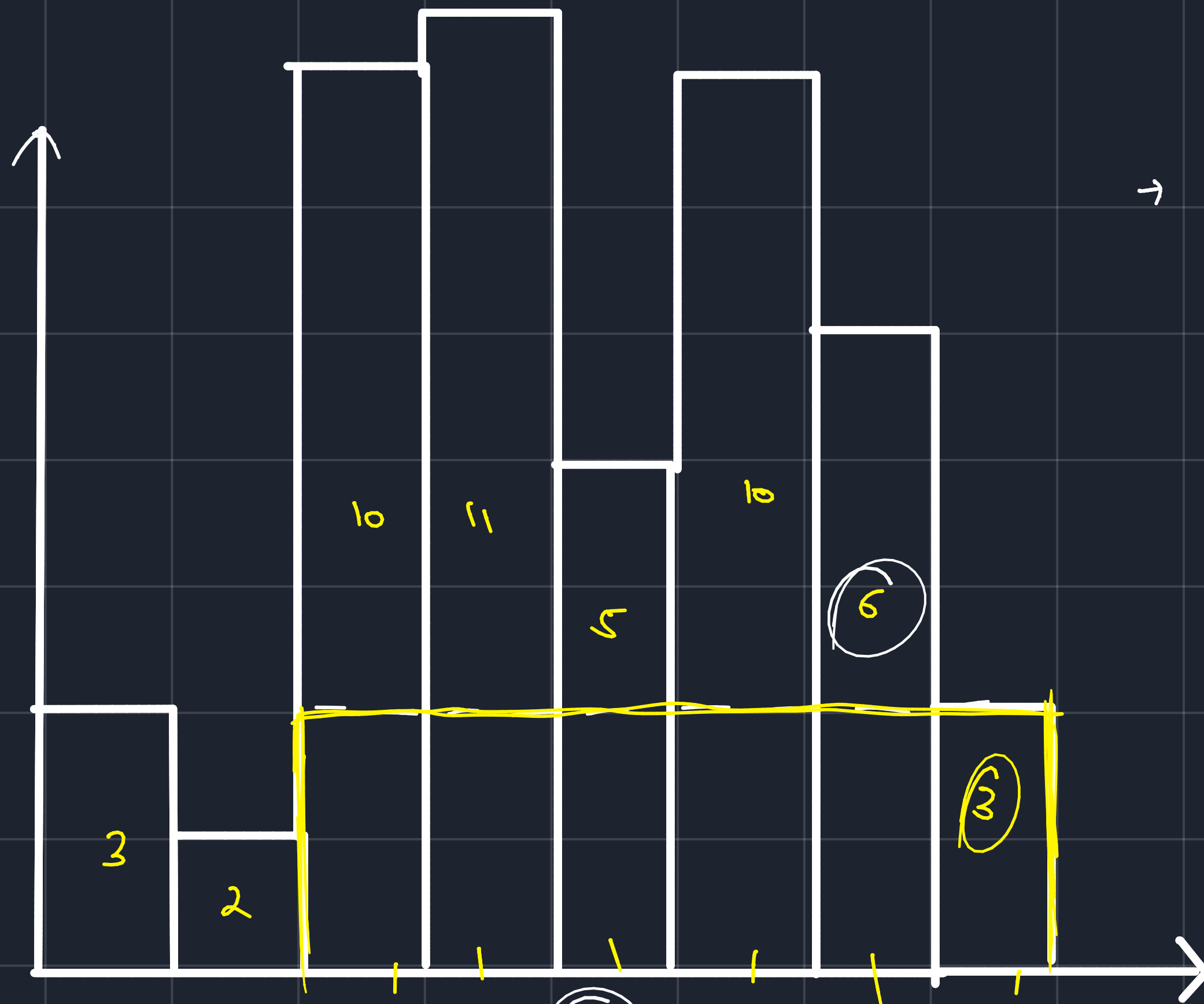
↳ we are storing

p-e-r in a stack.

↳ see if we observe.

↳ for the elements in the stack,

if the curr elem is smaller than
then, → it is the n-se to elem.



0 1 2 3 4 5 6 7
 ↑ ↑ ↑ ↑ ↑ ↑ ↑
 3 2 10 11 5 10 6 3

need $2 < 3$
 need p.s.e for (2)
 ↳ parent (3)

but for (3) → n.se is (2)
 ↳ p.s.e → element beneath in the stack

→ at index 4 → val 5

↳ for all elements before that are greater than

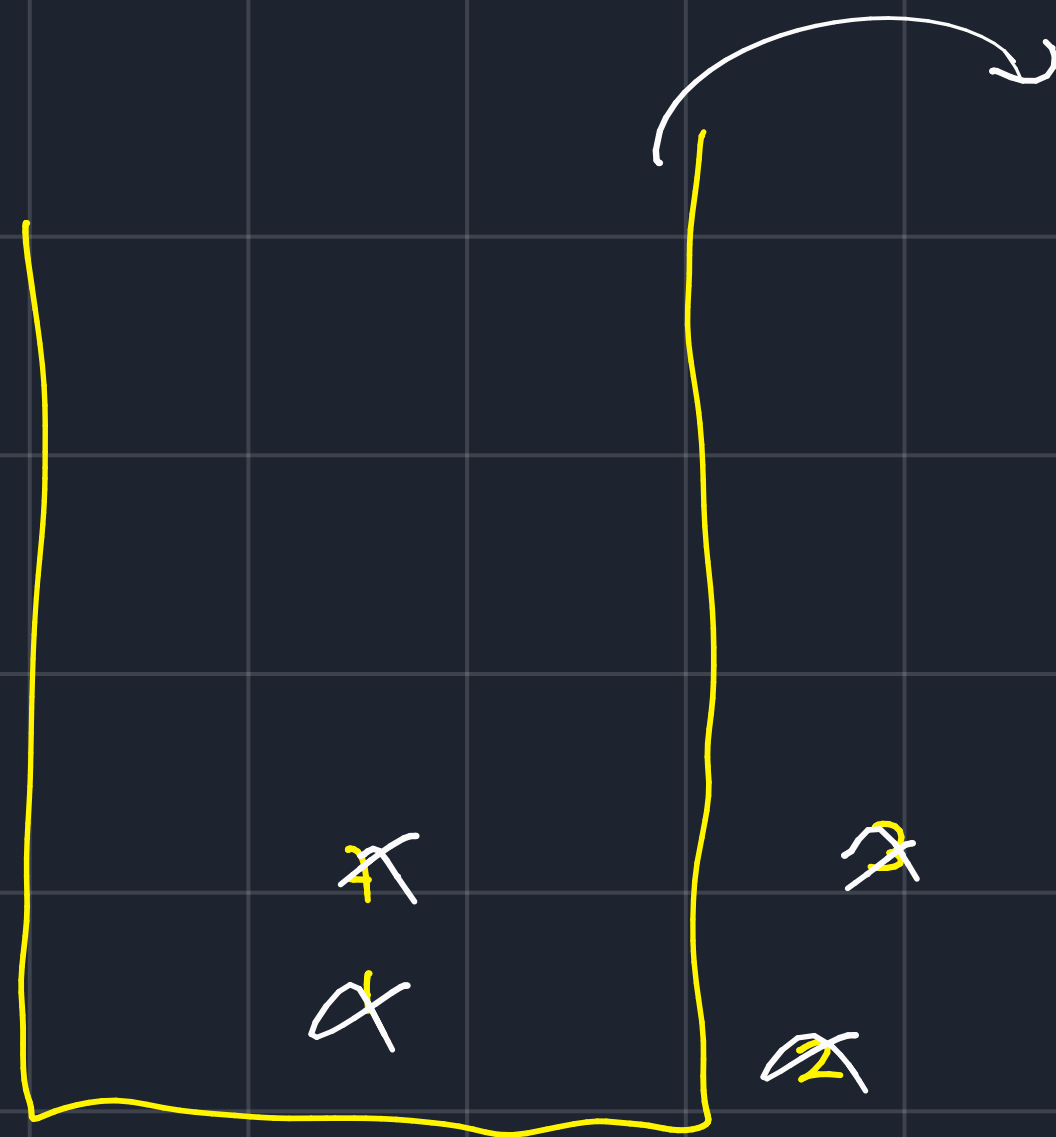
this → this is key n.se.

↳ while popping calculate the area

for → 11 → other n.se → 5
 → p.s.e → 11

ind
 7
 6
 5
 4
 3
 2
 1
 0

val
 3
 6 → $6 \times (7 - 5 - 1) = 6$
 10 → $10 \times (6 - 4 - 1) = 10$
 5 → $5 \times (7 - 1 - 1) = 25$
 11 → $11 \times (4 - 2 - 1) = 11$
 10 → $10 \times (4 - 1 - 1) = 10$



elements that are left in the
stack, means there is no
n.s for them.

Take imaginary index.

$$\rightarrow 3 \times (8 - 1 - 1) = \underline{10}$$

$$\rightarrow 2 \times (8 - (1)1) =$$

$$\rightarrow 2 \times (8) = \underline{16}$$

$$3 \times (1 - 1 - 1)$$

$$\underline{3 \times (1)}$$

